

| Category     | Function                        | Description                                                                                          | Syntax                                                                                   | Example of function                                                            | Possible result                                 |
|--------------|---------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------|
| Mathematical | <a href="#">add</a>             | Return the result from adding two numbers.                                                           | add(<summand_1>, <summand_2>)                                                            | add(1, 1.5)                                                                    | 2.5                                             |
| Dates        | <a href="#">addDays</a>         | Add a number of days to a timestamp.                                                                 | addDays('<timestamp>', <days>, '<format>?')                                              | addDays('2018-03-15T13:00:00Z', 10)                                            | "2018-03-25T00:00:000000Z"                      |
| Dates        | <a href="#">addHours</a>        | Add a number of hours to a timestamp.                                                                | addHours('<timestamp>', <hours>, '<format>?')                                            | addHours('2018-03-15T00:00:00Z', 10)                                           | "2018-03-15T10:00:000000Z"                      |
| Dates        | <a href="#">addMinutes</a>      | Add a number of minutes to a timestamp.                                                              | addMinutes('<timestamp>', <minutes>, '<format>?')                                        | addMinutes('2018-03-15T00:10:00Z', 10)                                         | "2018-03-15T00:20:00.000000Z"                   |
| Dates        | <a href="#">addSeconds</a>      | Add a number of seconds to a timestamp.                                                              | addSeconds('<timestamp>', <seconds>, '<format>?')                                        | addSeconds('2018-03-15T00:00:00Z', 10)                                         | "2018-03-15T00:00:10.000000Z"                   |
| Dates        | <a href="#">addToTime</a>       | <a href="#">Add a number of time units to a timestamp. See also getFutureTime.</a>                   | addToTime('<timestamp>', <interval>, '<timeUnit>', '<format>?')                          | addToTime('2018-01-01T00:00:00Z', 1, 'Day')                                    | "2018-01-02T00:00:00.000000Z"                   |
| Logical      | <a href="#">and</a>             | Check whether all expressions are true.                                                              | and(<expression1>, <expression2>)                                                        | and(false, true)                                                               | false                                           |
| Conversion   | <a href="#">array</a>           | <a href="#">Return an array from a single specified input. For multiple inputs, see createArray.</a> | array('<value>')                                                                         | array('hello')                                                                 | ["hello"]                                       |
| Conversion   | <a href="#">base64</a>          | Return the base64-encoded version for a string.                                                      | base64('<value>')                                                                        | base64('hello')                                                                | "aGVsbG8="                                      |
| Conversion   | <a href="#">base64ToBinary</a>  | Return the binary version for a base64-encoded string.                                               | base64ToBinary('<value>')                                                                | base64ToBinary('aGVsbG8=')                                                     | "01100001010001110101..."                       |
| Conversion   | <a href="#">base64ToString</a>  | Return the string version for a base64-encoded string.                                               | base64ToString('<value>')                                                                | base64ToString('aGVsbG8=')                                                     | "hello"                                         |
| Conversion   | <a href="#">binary</a>          | Return the binary version for an input value.                                                        | binary('<value>')                                                                        | binary('hello')                                                                | "0110100001100101011011..."                     |
| Conversion   | <a href="#">bool</a>            | Return the Boolean version for an input value.                                                       | bool(<value>)                                                                            | bool(1)                                                                        | true                                            |
| Conversion   | <a href="#">coalesce</a>        | Return the first non-null value from one or more parameters.                                         | coalesce(<object_1>, <object_2>, ...)                                                    | coalesce(null, true, false)                                                    | true                                            |
| Strings      | <a href="#">concat</a>          | Combine two or more strings, and return the combined string.                                         | concat('<text1>', '<text2>', ...)                                                        | concat('Hello', 'World')                                                       | "HelloWorld"                                    |
| Collection   | <a href="#">contains</a>        | Check whether a collection has a specific item.                                                      | contains('<collection>', '<value>')<br>OR contains([<collection>], '<value>')            | contains('hello world', 'world')                                               | true                                            |
| Dates        | <a href="#">convertFromUtc</a>  | Convert a timestamp from Universal Time Coordinated (UTC) to the target time zone.                   | convertFromUtc('<timestamp>', '<destinationTimeZone>', '<format>?')                      | convertFromUtc('2018-01-01T08:00:00.000000Z', 'Pacific Standard Time')         | "2018-01-01T00:00:00Z"                          |
| Dates        | <a href="#">convertTimeZone</a> | Convert a timestamp from the source time zone to the target time zone.                               | convertTimeZone('<timestamp>', '<sourceTimeZone>', '<destinationTimeZone>', '<format>?') | convertTimeZone('2018-01-01T08:00:00.000000Z', 'UTC', 'Pacific Standard Time') | "2018-01-01T00:00:00.000000Z"                   |
| Dates        | <a href="#">convertToUtc</a>    | Convert a timestamp from the source time zone to Universal Time Coordinated (UTC).                   | convertToUtc('<timestamp>', '<sourceTimeZone>', '<format>?')                             | convertToUtc('01/01/2018 00:00:00', 'Pacific Standard Time')                   | "2018-01-01T08:00:00.000000Z"                   |
| Conversion   | <a href="#">createArray</a>     | Return an array from multiple inputs.                                                                | createArray('<object1>', '<object2>', ...)                                               | createArray('h', 'e', 'l', 'l', 'o')                                           | ["h", "e", "l", "l", "o"]                       |
| Conversion   | <a href="#">dataUri</a>         | Return the data URI for an input value.                                                              | dataUri('<value>')                                                                       | dataUri('hello')                                                               | "data:text/plain;charset=utf-8;base64,aGVsbG8=" |
| Conversion   | <a href="#">dataUriToBinary</a> | Return the binary version for a data URI.                                                            | dataUriToBinary('<value>')                                                               | dataUriToBinary('data:text/plain;charset=utf-8;base64,aGVsbG8=')               | "011001000110000101110100011..."                |
| Conversion   | <a href="#">dataUriToString</a> | Return the string version for a data URI.                                                            | dataUriToString('<value>')                                                               | dataUriToString('data:text/plain;charset=utf-8;base64,aGVsbG8=')               | "hello"                                         |
| Dates        | <a href="#">dayOfMonth</a>      | Return the day of the month component from a timestamp.                                              | dayOfMonth('<timestamp>')                                                                | dayOfMonth('2018-03-15T13:27:36Z')                                             | 15                                              |

| Category     | Function                           | Description                                                                                             | Syntax                                                                                | Example of function                                                                     | Possible result                          |
|--------------|------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------|
| Dates        | <a href="#">dayOfWeek</a>          | Return the day of the week component from a timestamp.                                                  | dayOfWeek('<timestamp>')                                                              | dayOfWeek('2018-03-15T13:27:36Z')                                                       | 3                                        |
| Dates        | <a href="#">dayOfYear</a>          | Return the day of the year component from a timestamp.                                                  | dayOfYear('<timestamp>')                                                              | dayOfYear('2018-03-15T13:27:36Z')                                                       | 74                                       |
| Conversion   | <a href="#">decodeBase64</a>       | Return the string version for a base64-encoded string.                                                  | decodeBase64('<value>')                                                               | decodeBase64('aGVsbG8=')                                                                | "hello"                                  |
| Conversion   | <a href="#">decodeDataUri</a>      | Return the binary version for a data URI.                                                               | decodeDataUri('<value>')                                                              | decodeDataUri('data:text/plain;charset=utf-8;base64,aGVsbG8=')                          | "011001000110000101110100..."            |
| Conversion   | <a href="#">decodeUriComponent</a> | Return a string that replaces escape characters with decoded versions.                                  | decodeUriComponent('<value>')                                                         | decodeUriComponent('http%3A%2F%2Fcontoso.com')                                          | "https://contoso.com"                    |
| Mathematical | <a href="#">div</a>                | Return the result from dividing two numbers.                                                            | div(<dividend>, <divisor>)                                                            | div(10, 5)                                                                              | 2                                        |
| Collection   | <a href="#">empty</a>              | Check whether a collection is empty.                                                                    | empty('<collection>') OR<br>empty([<collection>])                                     | empty('abc')                                                                            | true                                     |
| Conversion   | <a href="#">encodeUriComponent</a> | Return a string that replaces URL-unsafe characters with escape characters.                             | encodeUriComponent('<value>')                                                         | encodeUriComponent('https://contoso.com')                                               | "http%3A%2F%2Fcontoso.com"               |
| Strings      | <a href="#">endsWith</a>           | Check whether a string ends with the specified substring.                                               | endsWith('<text>', '<searchText>')                                                    | endsWith('hello world', 'world')                                                        | true                                     |
| Logical      | <a href="#">equals</a>             | Check whether both values are equivalent.                                                               | equals('<object1>', '<object2>')                                                      | equals(true, 1)                                                                         | true                                     |
| Collection   | <a href="#">first</a>              | Return the first item from a collection.                                                                | first('<collection>') OR<br>first([<collection>])                                     | first('hello')                                                                          | "h"                                      |
| Conversion   | <a href="#">float</a>              | Return a floating point number for an input value.                                                      | float('<value>')                                                                      | float('10.333')                                                                         | 10.333                                   |
| Dates        | <a href="#">formatDateTime</a>     | Return the timestamp as a string in optional format.                                                    | formatDateTime('<timestamp>', '<format>?')                                            | formatDateTime('03/15/2018 12:00:00', 'yyyy-MM-ddTHH:mm:ss')                            | "2018-03-15T12:00:00"                    |
| Dates        | <a href="#">getFutureTime</a>      | <a href="#">Return the current timestamp plus the specified time units. See also addToTime.</a>         | getFutureTime(<interval>, <timeUnit>, <format>?)                                      | getFutureTime(5, 'Day')                                                                 | "2018-03-06T00:00:00.0000000Z"           |
| Dates        | <a href="#">getPastTime</a>        | <a href="#">Return the current timestamp minus the specified time units. See also subtractFromTime.</a> | getPastTime(<interval>, <timeUnit>, <format>?)                                        | getPastTime(5, 'Day')                                                                   | "2018-01-27T00:00:00.0000000Z"           |
| Logical      | <a href="#">greater</a>            | Check whether the first value is greater than the second value.                                         | greater(<value>, <compareTo>) OR<br>greater('<value>', '<compareTo>')                 | greater(10, 5)                                                                          | true                                     |
| Logical      | <a href="#">greaterOrEquals</a>    | Check whether the first value is greater than or equal to the second value.                             | greaterOrEquals(<value>, <compareTo>) OR<br>greaterOrEquals('<value>', '<compareTo>') | greaterOrEquals(5, 5)                                                                   | true                                     |
| Strings      | <a href="#">guid</a>               | Generate a globally unique identifier (GUID) as a string.                                               | guid()                                                                                | guid('P')                                                                               | "(c2ecc88d-88c8-4096-912c-d6f2e2b138ce)" |
| Logical      | <a href="#">if</a>                 | Check whether an expression is true or false. Based on the result, return a specified value.            | if(<expression>, <valueIfTrue>, <valueIfFalse>)                                       | if(equals(1, 1), 'yes', 'no')                                                           | "yes"                                    |
| Strings      | <a href="#">indexOf</a>            | Return the starting position for a substring.                                                           | indexOf('<text>', '<searchText>')                                                     | indexOf('hello world', 'world')                                                         | 6                                        |
| Conversion   | <a href="#">int</a>                | Return the integer version for a string.                                                                | int('<value>')                                                                        | int('10')                                                                               | 10                                       |
| Collection   | <a href="#">intersection</a>       | Return a collection that has <i>only</i> the common items across the specified collections.             | intersection([<collection1>], [<collection2>], ...)                                   | intersection(createArray(1, 2, 3), createArray(101, 2, 1, 10), createArray(6, 8, 1, 2)) | [1, 2]                                   |
| Collection   | <a href="#">join</a>               | Return a string that has <i>all</i> the items from an array, separated by the specified character.      | join([<collection>], '<delimiter>')                                                   | join(createArray('a', 'b', 'c'), ',')                                                   | "a.b.c"                                  |
| Conversion   | json                               | Return the JavaScript Object Notation (JSON) type value or object for a string or XML.                  | json('<value>')                                                                       | json('[1, 2, 3]')                                                                       | [1, 2, 3]                                |
| Collection   | <a href="#">last</a>               | Return the last item from a collection.                                                                 | last('<collection>') OR<br>last([<collection>])                                       | last('abcd')                                                                            | "d"                                      |

| Category     | Function                         | Description                                                                                                                                          | Syntax                                                                 | Example of function                                | Possible result                |
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| Strings      | <a href="#">lastIndexOf</a>      | Return the starting position for the last occurrence of a substring.                                                                                 | lastIndexOf('<text>', '<searchText>')                                  | lastIndexOf('hello world', 'world')                | 6                              |
| Collection   | <a href="#">length</a>           | Return the number of items in a string or array.                                                                                                     | length('<collection>') OR<br>length([<collection>])                    | length('abcd')                                     | 4                              |
| Logical      | <a href="#">less</a>             | Check whether the first value is less than the second value.                                                                                         | less(<value>, <compareTo>)                                             | less(5, 10)                                        | true                           |
| Logical      | <a href="#">lessOrEquals</a>     | Check whether the first value is less than or equal to the second value.                                                                             | lessOrEquals(<value>, <compareTo>)                                     | lessOrEquals(10, 10)                               | true                           |
| Mathematical | <a href="#">max</a>              | Return the highest value from a set of numbers or an array.                                                                                          | max(<number1>, <number2>, ...)                                         | max(1, 2, 3)                                       | 3                              |
| Mathematical | <a href="#">min</a>              | Return the lowest value from a set of numbers or an array.                                                                                           | min(<number1>, <number2>, ...)                                         | min(1, 2, 3)                                       | 1                              |
| Mathematical | <a href="#">mod</a>              | Return the remainder from dividing two numbers.                                                                                                      | mod(<dividend>, <divisor>)                                             | mod(3, 2)                                          | 1                              |
| Mathematical | <a href="#">mul</a>              | Return the product from multiplying two numbers.                                                                                                     | mul(<multiplicand1>, <multiplicand2>)                                  | mul(1, 2)                                          | 2                              |
| Logical      | <a href="#">not</a>              | Check whether an expression is false.                                                                                                                | not(<expression>)                                                      | not(false)                                         | true                           |
| Logical      | <a href="#">or</a>               | Check whether at least one expression is true.                                                                                                       | or(<expression1>, <expression2>)                                       | or(true, false)                                    | true                           |
| Mathematical | <a href="#">rand</a>             | Return a random integer from a specified range.                                                                                                      | rand(<minValue>, <maxValue>)                                           | rand(1, 5)                                         | 4                              |
| Mathematical | <a href="#">range</a>            | Return an integer array that starts from a specified integer.                                                                                        | range(<startIndex>, <count>)                                           | range(1, 4)                                        | [1, 2, 3, 4]                   |
| Strings      | <a href="#">replace</a>          | Replace a substring with the specified string, and return the updated string.                                                                        | replace('<text>', '<oldText>', '<newText>')                            | replace('the old string', 'old', 'new')            | "the new string"               |
| Collection   | <a href="#">skip</a>             | Remove items from the front of a collection, and return <i>all the other</i> items.                                                                  | skip([<collection>], <count>)                                          | skip(createArray(0, 1, 2, 3), 1)                   | [1, 2, 3]                      |
| Strings      | <a href="#">split</a>            | Return an array that contains substrings, separated by commas, from a larger string based on a specified delimiter character in the original string. | split('<text>', '<delimiter>')                                         | split('a_b_c', '_')                                | ["a", "b", "c"]                |
| Dates        | <a href="#">startOfDay</a>       | Return the start of the day for a timestamp.                                                                                                         | startOfDay('<timestamp>', '<format>?')                                 | startOfDay('2018-03-15T13:30:30Z')                 | "2018-03-15T00:00:00.0000000Z" |
| Dates        | <a href="#">startOfHour</a>      | Return the start of the hour for a timestamp.                                                                                                        | startOfHour('<timestamp>', '<format>?')                                | startOfHour('2018-03-15T13:30:30Z')                | "2018-03-15T13:00:00.0000000Z" |
| Dates        | <a href="#">startOfMonth</a>     | Return the start of the month for a timestamp.                                                                                                       | startOfMonth('<timestamp>', '<format>?')                               | startOfMonth('2018-03-15T13:30:30Z')               | "2018-03-01T00:00:00.0000000Z" |
| Strings      | <a href="#">startsWith</a>       | Check whether a string starts with a specific substring.                                                                                             | startsWith('<text>', '<searchText>')                                   | startsWith('hello world', 'hello')                 | true                           |
| Conversion   | <a href="#">string</a>           | Return the string version for an input value.                                                                                                        | string(<value>)                                                        | string(10)                                         | "10"                           |
| Mathematical | <a href="#">sub</a>              | Return the result from subtracting the second number from the first number.                                                                          | sub(<minuend>, <subtrahend>)                                           | sub(10.3, .3)                                      | 10                             |
| Strings      | <a href="#">substring</a>        | Return characters from a string, starting from the specified position.                                                                               | substring('<text>', <startIndex>, <length>)                            | substring('hello world', 6, 5)                     | "world"                        |
| Dates        | <a href="#">subtractFromTime</a> | <a href="#">Subtract a number of time units from a timestamp. See also <a href="#">getPastTime</a>.</a>                                              | subtractFromTime('<timestamp>', <interval>, '<timeUnit>', '<format>?') | subtractFromTime('2018-01-02T00:00:00Z', 1, 'Day') | "2018-01-01T00:00:00.0000000Z" |
| Collection   | <a href="#">take</a>             | Return items from the front of a collection.                                                                                                         | take('<collection>', <count>) OR<br>take([<collection>], <count>)      | take('abcde', 3)                                   | "abc"                          |

| Category   | Function                             | Description                                                                                                                | Syntax                                                                                          | Example of function                                         | Possible result                |
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| Dates      | <a href="#">ticks</a>                | Return the ticks property value for a specified timestamp.                                                                 | ticks('<timestamp>')                                                                            | ticks('2018-03-15T13:30:30Z')                               | 636566556300000000             |
| Strings    | <a href="#">toLowerCase</a>          | Return a string in lowercase format.                                                                                       | toLowerCase('<text>')                                                                           | toLowerCase('Hello World')                                  | "hello world"                  |
| Strings    | <a href="#">toUpperCase</a>          | Return a string in uppercase format.                                                                                       | toUpperCase('<text>')                                                                           | toUpperCase('Hello World')                                  | "HELLO WORLD"                  |
| Strings    | <a href="#">trim</a>                 | Remove leading and trailing whitespace from a string, and return the updated string.                                       | trim('<text>')                                                                                  | trim(' Hello World ')                                       | "Hello World"                  |
| Collection | <a href="#">union</a>                | Return a collection that has <i>all</i> the items from the specified collections.                                          | union('<collection1>', '<collection2>', ...) OR<br>union([<collection1>], [<collection2>], ...) | union(createArray(1, 2, 3), createArray(1, 2, 10, 101))     | [1, 2, 3, 10, 101]             |
| Conversion | <a href="#">uriComponent</a>         | Return the URI-encoded version for an input value by replacing URL-unsafe characters with escape characters.               | uriComponent('<value>')                                                                         | uriComponent('https://contoso.com')                         | "http%3A%2F%2Fcontoso.com"     |
| Conversion | <a href="#">uriComponentToBinary</a> | Return the binary version for a URI-encoded string.                                                                        | uriComponentToBinary('<value>')                                                                 | uriComponentToBinary('http%3A%2F%2Fcontoso.com')            | "0010001001101000011101..."    |
| Conversion | <a href="#">uriComponentToString</a> | Return the string version for a URI-encoded string.                                                                        | uriComponentToString('<value>')                                                                 | uriComponentToString('http%3A%2F%2Fcontoso.com')            | "https://contoso.com"          |
| Dates      | <a href="#">utcNow</a>               | Return the current timestamp as a string.                                                                                  | utcNow('<format>')                                                                              | utcNow()                                                    | "2018-04-15T13:00:00.0000000Z" |
| Conversion | <a href="#">xml</a>                  | Return the XML version for a string.                                                                                       | xml('<value>')                                                                                  | xml(json('{ \"name\": \"Sophia Owen\" }'))                  | <name>Sophia Owen</name>       |
| Conversion | <a href="#">xpath</a>                | Check XML for nodes or values that match an XPath (XML Path Language) expression, and return the matching nodes or values. | xpath('<xml>', '<xpath>')                                                                       | xpath(xml(parameters('items')), 'sum(/produce/item/count)') | 30                             |